

JavaScript_Day-4_Hands_On—Vasundhara

Problem 1

Assessment Goal: Ensure learners understand responsiveness and screen adaptability.

Hands-on Tasks:

1. Add viewport meta tag to the HTML page
2. Use media queries to:
 - o Change background color on mobile screen
 - o Adjust font size for smaller screens
3. Convert navigation into vertical layout on mobile
4. Test the page using browser responsive mode

Expected Outcome:

A webpage that looks different and readable on mobile and desktop screens.

Source code:

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Home Page</title>
7
8   <style>
9     body {
10       background-color: #bisque;
11       font-family: 'Times New Roman', Times, serif;
12       font-size: 16px;
13       margin: 0;
14     }
15     h1 {
16       text-align: center;
17     }
18     a {
19       background-color: #goldenrod;
20       padding: 10px;
21     }
22     .A .B {
23       list-style: none;
24       display: flex;
25       justify-content: center;
26       padding: 0;
27       margin: 0;
28     }
29     .A .B .C {
```

```

29     .A .B .C {
30         margin: 0 15px;
31     }
32     .A .B .C a {
33         color: white;
34         text-decoration: none;
35     }
36     section {
37         padding: 20px;
38         text-align: center;
39     }
40
41     /* Mobile View */
42     @media (max-width: 576px) {
43         body {
44             background-color: teal;
45             font-size: 14px;
46         }
47         .A .B {
48             flex-direction: column;
49             align-items: center;
50         }
51         .A .B .C {
52             margin: 10px 0;
53         }
54     }

```

```

42     @media (max-width: 576px) {
51         .A .B .C {
52             margin: 10px 0;
53         }
54     }
55 </style>
56 </head>
57 <body>
58     <h1>HBK STORE</h1>
59     <nav class="A">
60         <ul class="B">
61             <li class="C"><a href="#home">Home</a></li>
62             <li class="C"><a href="#about">About</a></li>
63             <li class="C"><a href="#contact">Contact</a></li>
64         </ul>
65     </nav>
66     <section>
67         <h1>Welcome to our Page</h1>
68         <p>This page changes layout and style on mobile screens.</p>
69     </section>
70 </body>
71 </html>

```

Output:



Explanation:

This code creates a simple webpage for Store with a navigation bar and some basic styling using CSS. It also uses a media query to change the layout and background color when the screen size is small (like on a mobile). On a desktop, the menu appears in a horizontal line with a light background, but on mobile, the menu becomes vertical and the background color changes, making the page responsive and mobilefriendly.

Problem 2: Student Grade Evaluator (Level-1)

Scenario

A school wants a simple JavaScript program to evaluate a student's performance based on marks obtained in a subject.

📌 Requirements

- Accept the student's marks as a variable
- Use if-else statements to assign grades:
 - Marks ≥ 75 $\rightarrow$ Grade A
 - Marks ≥ 60 $\rightarrow$ Grade B

- Marks $\geq 40 \rightarrow$ Grade C
- Marks $< 40 \rightarrow$ Fail

Display the grade on the web page or console

Technical Constraints

- Use JavaScript variables (let or const)
- Use numeric data types
- Use comparison and logical operators
- No functions or arrays allowed
- Output using console.log() or document.write()

Learning Outcome

You should be able to:

- Declare and use variables
- Apply comparison operators
- Implement conditional logic using if-else
- Understand decision-making in JavaScript

Source code:

```
1  <!-- JS Day4 Hands on Problem Statement2 vasundhara -->
2  <!DOCTYPE html>
3  <html>
4  <head>
5      <meta charset="UTF-8">
6      <meta name="viewport" content="width=device-width, initial-scale=1.0">
7      <title>Student Grade Evaluator</title>
8
9      <style>
10         h1 {
11             text-align: center;
12         }
13         body {
14             background-color: pink;
15         }
16     </style>
17 </head>
18
19 <body>
20
21     <h1>Student Grade Evaluator</h1>
22
23     <script>
24         let score = prompt("Enter your Score:");
25         score = Number(score);
26
27         if (score >= 75) {
28             console.log("Grade A");
29         }
```

```
24 <script>
25 let score = prompt("Enter your Score:");
26 score = Number(score);
27
28 if (score >= 75) {
29   console.log("Grade A");
30   alert("Grade A");
31 }
32 else if (score >= 60) {
33   console.log("Grade B");
34   alert("Grade B");
35 }
36 else if (score >= 40) {
37   console.log("Grade C");
38   alert("Grade C");
39 }
40 else {
41   console.log("Fail");
42   alert("Fail");
43 }
44 </script>
45
46 </body>
47 </html>
```

Output:

127.0.0.1:5500 says

Enter your Score:

OK

Cancel

127.0.0.1:5500 says

Grade B

OK

Explanation: This code creates a simple Student Grade Evaluator page using JavaScript. It asks the user to enter their score through a prompt, checks the marks using if-else conditions, and then shows the grade (A, B, C, or Fail) in an alert box.

Problem 3: Simple Discount Calculator (Level-1)

Scenario

An online store wants to apply a discount based on the total purchase amount.

Requirements

- Store purchase amount in a variable
- Apply discount rules:
 - Amount $\geq 5000 \rightarrow$ 20% discount
 - Amount $\geq 3000 \rightarrow$ 10% discount
 - Amount $< 3000 \rightarrow$ No discount
- Calculate and display:
 - Discount amount
 - Final payable amount

Technical Constraints

- Use arithmetic operators
- Use if-else statements
- Use only primitive data types

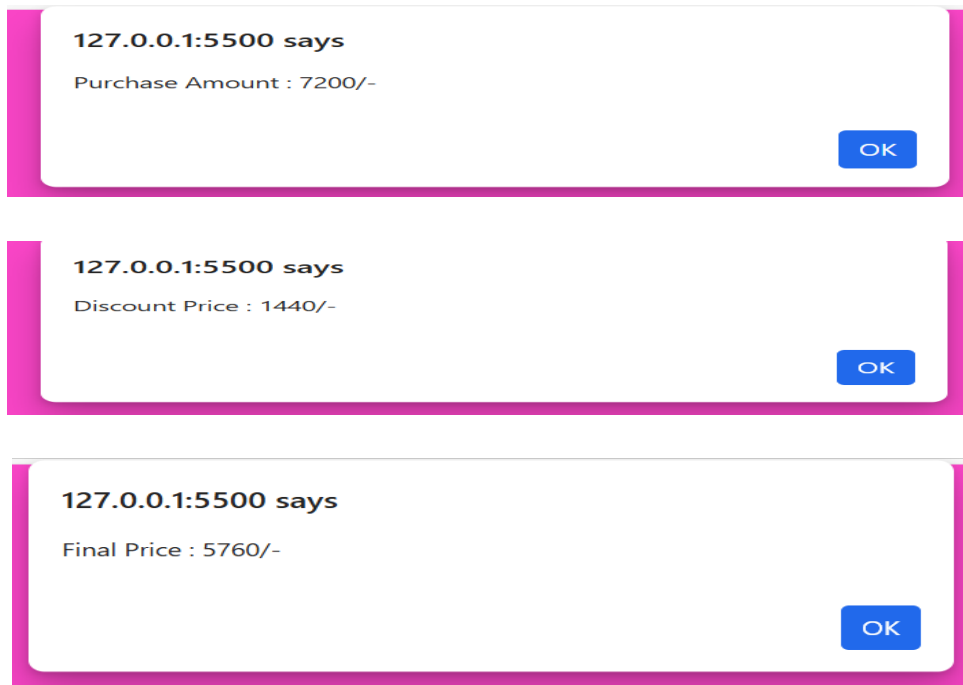
No user input (hardcoded values allowed)

Learning Outcome

You will be able to:

- Perform calculations using operators
- Work with expressions
- Apply conditional statements
- Build real-world logic using JavaScript basic

Output:



Explanation: This code creates a Simple Discount Calculator using JavaScript. It checks the purchase amount (₹7200) and applies a 20% discount if it's ₹5000 or more, then calculates the final price. It shows the purchase amount, the discount (₹1440), and the final price (₹5760) using alert messages.

Problem 4: Traffic Signal Simulator (Level-2)

Scenario

A traffic control system needs a JavaScript program that displays instructions based on traffic signal color.

Requirements

- Store signal color in a variable ("red", "yellow", "green")
- Use a **switch statement** to display:
- Red → Stop

- Yellow → Get Ready
- Green → Go

Handle invalid signal input gracefully

🔧 Technical Constraints

- Must use switch-case
- Use string data types
- Use console.log() for output
- No if-else allowed

🎯 Learning Outcome

Learners should be able to:

- Use switch statements effectively
- Compare string values
- Handle multiple conditions cleanly
- Understand control flow alternative

Source code:

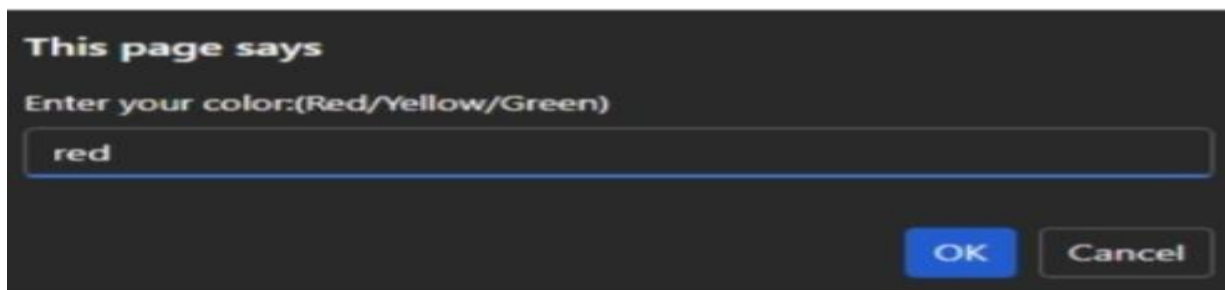
```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>Traffic Signal Simulator</title>
7
8      <style>
9          h1{
10             text-align: center;
11          }
12          body{
13             background-color: rgb(182, 40, 15);
14          }
15      </style>
16 </head>
17 <body>
18
19     <h1>Traffic Signal Simulator</h1>
20
21     <script>
22         let color = prompt("Enter your color:(Red/Yellow/Green)");
23
24         switch (color) {
25             case "red":
26                 console.log("Stop");
27                 break;
28             case "yellow":
```

```

21     <script>
24         switch (color) {
25             case "red":
26                 console.log("Stop");
27                 break;
28
29             case "yellow":
30                 console.log("Get Ready");
31                 break;
32
33             case "green":
34                 console.log("Go");
35                 break;
36
37             default:
38                 console.log("Invalid Color");
39         }
40     </script>
41
42 </body>
43 </html>

```

Output:



This page says

Enter your color:(Red/Yellow/Green)

OK Cancel



Stop index.html:25

This page says

Enter your color:(Red/Yellow/Green)

OK Cancel

Get Ready index.html:29

This page says

Enter your color:(Red/Yellow/Green)

OK Cancel

Go index.html:33

This page says

Enter your color:(Red/Yellow/Green)

OK Cancel

Explanation:

Explanation: This code creates a simple Traffic Signal Simulator using JavaScript. It asks the user to enter a color (red, yellow, or green) and uses a switch statement to display the corresponding action like Stop, Get Ready, or Go in the console.

Problem 5: Number Analysis Tool (Level-2)

Scenario

A utility program is required to analyze numbers and provide insights such as positivity, parity, and range.

Requirements

- Store a number in a variable
- Use **conditional (ternary) operator** to check:
 - Positive or Negative
- Use **if-else** to check:
 - Even or Odd
- Use a **loop** to print all numbers from 1 to the given number

Technical Constraints

- Store a number in a variable
- Use conditional (ternary) operator to check:
 - Positive or Negative
- Use if-else to check:
 - Even or Odd
- Use a loop to print all numbers from 1 to the given number

Learning Outcome

You will be able to:

- Combine multiple control flow techniques
- Use loops for iteration

Source code:

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8">
5    <meta name="viewport" content="width=device-width, initial-scale=1.0">
6    <title>Number Analysis Tool</title>
7
8    <style>
9      h1 {
10        text-align: center;
11      }
12
13      body {
14        background-color: #beige;
15        font-family: Arial, sans-serif;
16      }
17    </style>
18  </head>
19  <body>
20
21    <h1>Number Analysis Tool</h1>
22
23    <script>
24      let number = prompt("Enter any Number:");
25      number = Number(number);
26
27      let result = (number >= 0) ? "Positive" : "Negative";
28      console.log("Number is:", result);
29
```

```
21    <h1>Number Analysis Tool</h1>
22
23    <script>
24      let number = prompt("Enter any Number:");
25      number = Number(number);
26
27      let result = (number >= 0) ? "Positive" : "Negative";
28      console.log("Number is:", result);
29
30      if (number % 2 === 0) {
31        console.log("Number is Even");
32      } else {
33        console.log("Number is Odd");
34      }
35
36      // Print numbers from 1 to given number
37      console.log("Numbers from 1 to " + number + ":");
38      for (let i = 1; i <= number; i++) {
39        console.log(i);
40      }
41    </script>
42
43  </body>
44  </html>
```

output:

output:

This page says

Enter any Number:

Number is Positive	index.html:25
Number is Odd	index.html:30
Numbers from 1 to 11:	index.html:33
1	index.html:36
2	index.html:36
3	index.html:36
4	index.html:36
5	index.html:36
6	index.html:36
7	index.html:36
8	index.html:36
9	index.html:36
10	index.html:36
11	index.html:36

Explanation:

This code creates a Number Analysis Tool using JavaScript. It asks the user to enter a number, then checks whether it is positive or negative and whether it is even or odd. After that, it prints all numbers from 1 up to the entered number in the console.